



WCP Series

Industrial Process Water Chillers

WCP chillers keep process water at a prescribed set temperature using refrigeration. Each unit is self-contained, with a closed stainless steel tank, high-pressure pump and air- or water-cooled condenser, and is factory tested before delivery.

Capacity

0.5 to 20 TR

Temperature difference

5 °C

Inlet water pressure

2 to 4 bar g

Inlet water temperature

5 to 30 °C

Win Equipments

SF No. 4/195 B, Kallangadu, Nadu Arasur, Arasur Post, Coimbatore 641
407, Tamil Nadu, India
+91 95972 28969 · +91 95972 28975 · info@winequipments.com



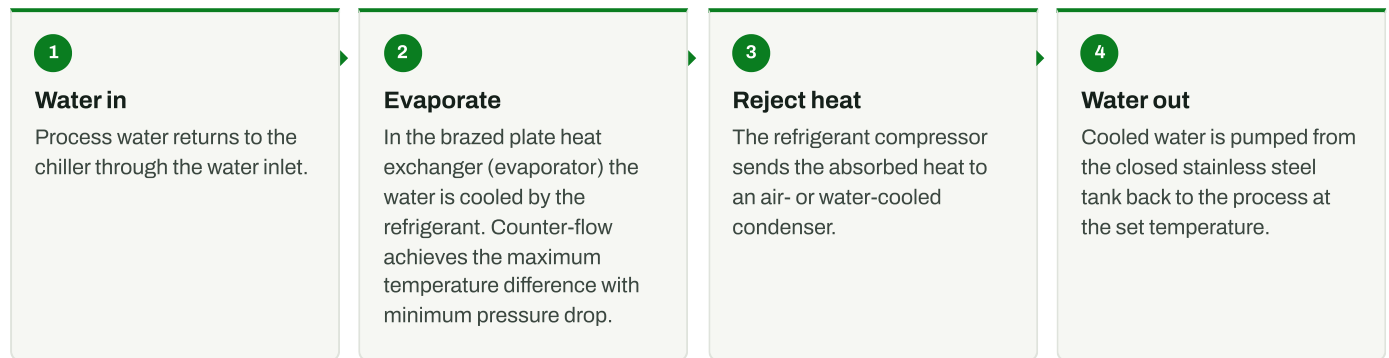
Full model range,
photos and
sizing tools



WORKING PRINCIPLE

How it works

The chilling plant has two circuits: a refrigeration system (compressor, condenser, fan, expansion device and evaporator) and a water circulating system (pump, motor and tubing).



CONSTRUCTION

Features

Brazed plate heat exchanger

- Sized to keep fluid velocities and turbulence high
- Counter-flow for maximum temperature difference
- Minimum pressure drop

Tank

- Non-corrosive insulated stainless steel tank
- Closed tank: no heat gain
- Removable steel covers for access

Refrigeration compressor

- Hermetically sealed reciprocating or scroll
- Suitable for eco-friendly gases
- Low noise, low power consumption

Controls

- Programmed temperature controller with alarms
- On-delay protects the compressor
- High / low pressure gauges and switches

Circulating pump

- Stainless steel impeller, mechanical seal
- Rated for 24-hour duty
- Motor overload protection

Hot gas bypass valve

- Prevents evaporator temperature dropping below 5 °C
- Manual or automatic

TECHNICAL SPECIFICATIONS

WCP Series models

Rating conditions

Water outlet temperature (ideal) 15 °C	Ambient temperature (ideal) 40 °C	Water ΔT 5 °C	Water inlet pressure 1 bar g ideal · 2–4 bar g max
--	---	-------------------------	--

Model	Capacity TR	Capacity m³/hr	Refrigeration compressor kW	Electrical V / Ph / Hz	Cooling fans qty	Condenser	Water flow LPM	Water pump HP	Water tank litres	Tank material	Shipping weight kg	End connection BSP / NB
WCP 005	0.5	0.3	0.9	230 / 1 / 50	1	Air / Water	6	0.5	15	SS	125	¾"
WCP 007	0.75	0.45	1.2	230 / 1 / 50	1	Air / Water	8	0.5	20	SS	135	¾"
WCP 010	1	0.6	1.5	230 / 1 / 50	1	Air / Water	10	0.5	25	SS	150	¾"
WCP 020	2	1.2	2.5	230 / 1 / 50	1	Air / Water	21	0.75	40	SS	200	1"
WCP 030	3	1.8	3.4	440 / 3 / 50	1	Air / Water	32	1	60	SS	240	1"
WCP 050	5	3.0	6.0	440 / 3 / 50	2	Air / Water	52	1.5	80	SS	350	1½"
WCP 075	7.5	4.5	9.2	440 / 3 / 50	3	Air / Water	80	2	140	SS	475	1½"
WCP 100	10	6.0	11.5	440 / 3 / 50	4	Air / Water	100	2	180	SS / MS	550	2"
WCP 150	15	9.0	18.0	440 / 3 / 50	6	Air / Water	155	2	250	SS / MS	700	2"
WCP 200	20	12.0	22.5	440 / 3 / 50	8	Air / Water	202	2	320	SS / MS	1000	2"

- › Model code: WCP + capacity in TR (e.g. WCP 050 = 5 TR). Working pressure 2–4 bar g max.
- › Pressure drop within 0.2 bar g max.
- › Rated power is the maximum power consumed at rated conditions.
- › Voltage range 180–260 V for 1 Ph and 380–420 V for 3 Ph.

Sizing for your conditions

Heat load $Q = m \times \Delta T$ (m = mass flow in LPM, ΔT = inlet – outlet temperature). Q in kcal/min $\times 60$ = kcal/hr.

Water outlet temperature

°C	5	10	15	20	25
Factor	0.6	0.75	1.0	1.16	1.3

Ambient temperature

°C	30	35	40	45	50
Factor	1.2	1.1	1.0	0.9	0.8

APPLICATIONS

Automobile

Injection moulding & plastics

Hydraulic presses & drives

Compressor cooling

Milk, dairy & pasteurising

Food products

Lubricant & oil cooling

Pulp & paper

Printing ink & paper waxing

Sugar

Footwear

FROM OUR WORKS



Process chiller with WIN control panel and pressure gauges



Large process chiller, blue frame



Large process chiller, control panel side



Large chiller in assembly, side panels off

QUESTIONS

What two values do I need to select a chiller?

The required water outlet (cold) temperature, and the heat load to be removed. Heat load Q = mass flow (LPM) $\times \Delta T$ between inlet and outlet temperature.

What is the tank made of?

Up to WCP 075 the tank is stainless steel. WCP 100 to WCP 200 are offered in stainless or mild steel.

Are WCP chillers air cooled or water cooled?

Every WCP model is available with either an air-cooled or a water-cooled condenser.

How does ambient temperature affect capacity?

Capacity is rated at 40 °C ambient. At 45 °C apply a factor of 0.9, at 50 °C 0.8.

Talk to an engineer

Send your compressor capacity, heat load or water flow, operating pressure, temperatures and site conditions. We reply with a recommended model and quotation.

Win Equipments

SF No. 4/195 B, Kallangadu, Nadu Arasur
Arasur Post, Coimbatore 641 407
Tamil Nadu, India

+91 95972 28969 · +91 95972 28975

WhatsApp +91 95972 28969

info@winequipments.com ·

winequipments.com

